



# Bank Churn Prediction

## Group 3



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# Predicting Credit Card Customer Churn

## Project Background

- Customer churn reduces profits; retaining customers is cheaper than acquiring new ones.
- Data science helps spot churn patterns early (e.g., age, income, card usage).
- Banks can use insights to offer targeted promotions and improve loyalty.

## Our Project

- Analyze customer data
- Identify churn factors

## Project Goal:

Develop a machine learning model to predict credit card churn.



## Objectives:

1. Select key financial & demographic features
2. Analyze churn-driving variables
3. Achieve  $\geq 80\%$  model accuracy
4. Recommend actions to retain at-risk customer



## Outcome:

Support banks with data-driven retention strategies.

# Overview of the Dataset

1.Contains **10,000 rows** and **23 columns**

2. Includes **customer banking information:**

- Demographics
- Credit card usage patterns
- Financial behaviors

3. **Key features:** Customer Age, Income Category, Total Transaction Amount, Credit Limit, Card Category

4. **Purpose:** To perform predictive analysis on customer attrition

```
CustomerID CustomerAge CustomerStatus Age Income_Cat Card_Cat \
2 713982108 Existing Customer 51 M 3
3 769911858 Existing Customer 40 F 4
4 709106358 Existing Customer 40 M 3

Education_Level Marital_Status Income_Category Card_Category \
0 High School Married $60K - $80K Blue
1 Graduate Single Less than $40K Blue
2 Graduate Married $80K - $120K Blue
3 High School Unknown Less than $40K Blue
4 Uneducated Married $60K - $80K Blue

Months_on_book ... Months_Inactive_12_mon Contacts_Count_12_mon \
0 39 ... 1 3
1 44 ... 1 2
2 36 ... 1 0
3 34 ... 4 1
4 21 ... 1 0

Credit_Limit Total_Revolving_Bal Avg_Open_To_Buy Total_Amt_Chng_Q4_Q1 \
0 12691.0 777 11914.0 1.335
1 8256.0 864 7392.0 1.541
2 3418.0 0 3418.0 2.594
3 3313.0 2517 796.0 1.405
4 4716.0 0 4716.0 2.175

Total_Trans_Amt Total_Trans_Ct Total_Ct_Chng_Q4_Q1 Avg_Utilization_Ratio \
0 10124 10500083 Attrited Customer 44 F 1
10125 717406983 Attrited Customer 30 M 2
10126 714337233 Attrited Customer 43 F 2

Education_Level Marital_Status Income_Category Card_Category \
10122 Graduate Single $40K - $60K Blue
10123 Unknown Divorced $40K - $60K Blue
10124 High School Married Less than $40K Blue
10125 Graduate Unknown $40K - $60K Blue
10126 Graduate Married Less than $40K Silver

Months_on_book ... Months_Inactive_12_mon Contacts_Count_12_mon \
10122 40 ... 2 3
10123 25 ... 2 3
10124 36 ... 3 4
10125 36 ... 3 3
10126 25 ... 2 4

Credit_Limit Total_Revolving_Bal Avg_Open_To_Buy \
10122 4003.0 1851 2152.0
10123 4277.0 2186 2091.0
10124 5409.0 0 5409.0
10125 5281.0 0 5281.0
10126 10388.0 1961 8427.0

Total_Amt_Chng_Q4_Q1 Total_Trans_Amt Total_Trans_Ct \
10122 0.703 15476 117
10123 0.804 8764 69
```

Code output of dataset overview

# High Level Statistics

Basic summary metrics (like mean, median, standard deviation, and frequency counts)that provide a snapshot of the dataset.

**Purpose:**

- This step helps us quickly understand the structure, distributions, and potential issues in the data before diving deeper. It’s useful for spotting trends, anomalies, and early insights.

**Application in Our Project:**

We applied high-level statistics to both numerical and categorical features. This allowed us to:

- Summarise customer demographics and behavior
- Identify early signs of imbalance or unusual data points

| Statistical summary of features: |              |              |                 |                  |
|----------------------------------|--------------|--------------|-----------------|------------------|
|                                  | CLIENTNUM    | Customer_Age | Dependent_count | Months_on_book \ |
| count                            | 1.012700e+04 | 10127.000000 | 10127.000000    | 10127.000000     |
| mean                             | 7.391776e+08 | 46.325960    | 2.346203        | 35.928409        |
| std                              | 3.690378e+07 | 8.016814     | 1.298908        | 7.986416         |
| min                              | 7.080821e+08 | 26.000000    | 0.000000        | 13.000000        |
| 25%                              | 7.130368e+08 | 41.000000    | 1.000000        | 31.000000        |
| 50%                              | 7.179264e+08 | 46.000000    | 2.000000        | 36.000000        |
| 75%                              | 7.731435e+08 | 52.000000    | 3.000000        | 40.000000        |
| max                              | 8.283431e+08 | 73.000000    | 5.000000        | 56.000000        |

|       | Total_Relationship_Count | Months_Inactive_12_mon \ |
|-------|--------------------------|--------------------------|
| count | 10127.000000             | 10127.000000             |
| mean  | 3.812580                 | 2.341167                 |
| std   | 1.554408                 | 1.010622                 |
| min   | 1.000000                 | 0.000000                 |
| 25%   | 3.000000                 | 2.000000                 |
| 50%   | 4.000000                 | 2.000000                 |
| 75%   | 5.000000                 | 3.000000                 |
| max   | 6.000000                 | 6.000000                 |

|       | Contacts_Count_12_mon | Credit_Limit | Total_Revolving_Bal \ |
|-------|-----------------------|--------------|-----------------------|
| count | 10127.000000          | 10127.000000 | 10127.000000          |
| mean  | 2.455317              | 8631.953698  | 1162.814061           |
| std   | 1.106225              | 9088.776650  | 814.987335            |
| min   | 0.000000              | 1438.300000  | 0.000000              |
| 25%   | 2.000000              | 2555.000000  | 359.000000            |
| 50%   | 2.000000              | 4549.000000  | 1276.000000           |
| 75%   | 3.000000              | 11067.500000 | 1784.000000           |
| max   | 6.000000              | 34516.000000 | 2517.000000           |

|       | Avg_Open_To_Buy | Total_Amt_Chng_Q4_Q1 | Total_Trans_Amt | Total_Trans_Ct \ |
|-------|-----------------|----------------------|-----------------|------------------|
| count | 10127.000000    | 10127.000000         | 10127.000000    | 10127.000000     |
| mean  | 7469.139637     | 0.759941             | 4404.086304     | 64.858695        |
| std   | 9090.685324     | 0.219207             | 3397.129254     | 23.472570        |
| min   | 3.000000        | 0.000000             | 510.000000      | 10.000000        |
| 25%   | 1324.500000     | 0.631000             | 2155.500000     | 45.000000        |
| 50%   | 3474.000000     | 0.736000             | 3899.000000     | 67.000000        |
| 75%   | 9859.000000     | 0.859000             | 4741.000000     | 81.000000        |
| max   | 34516.000000    | 3.397000             | 18484.000000    | 139.000000       |

|       | Total_Ct_Chng_Q4_Q1 | Avg_Utilization_Ratio |
|-------|---------------------|-----------------------|
| count | 10127.000000        | 10127.000000          |
| mean  | 0.712222            | 0.274894              |
| std   | 0.238086            | 0.275691              |
| min   | 0.000000            | 0.000000              |
| 25%   | 0.582000            | 0.023000              |
| 50%   | 0.702000            | 0.176000              |
| 75%   | 0.818000            | 0.503000              |
| max   | 3.714000            | 0.999000              |

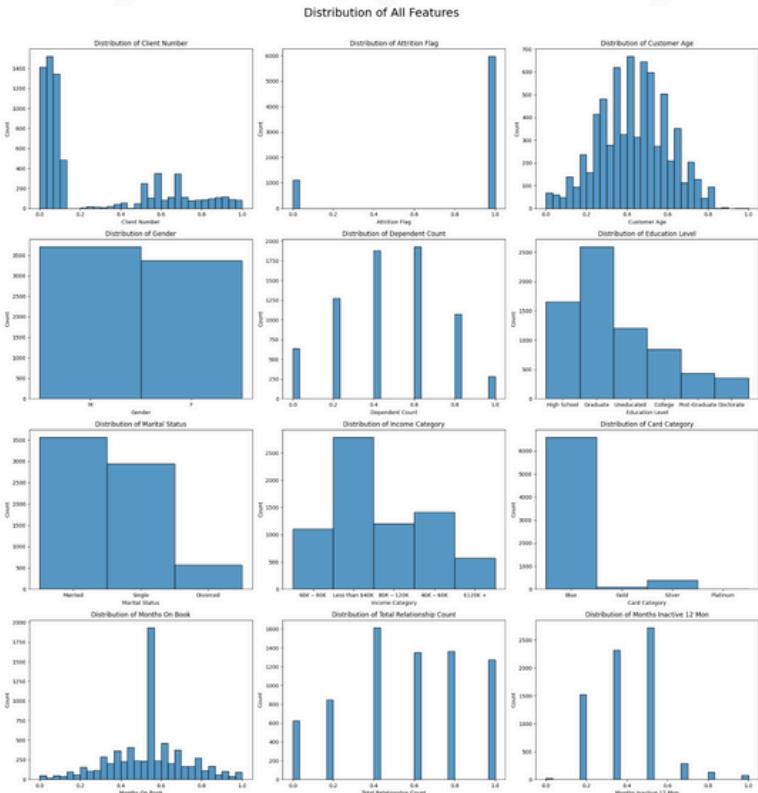


# Exploratory Data Analysis

**Purpose**

Forms the foundation for deeper analysis and ensures that predictive models are built on well-understood, reliable data.

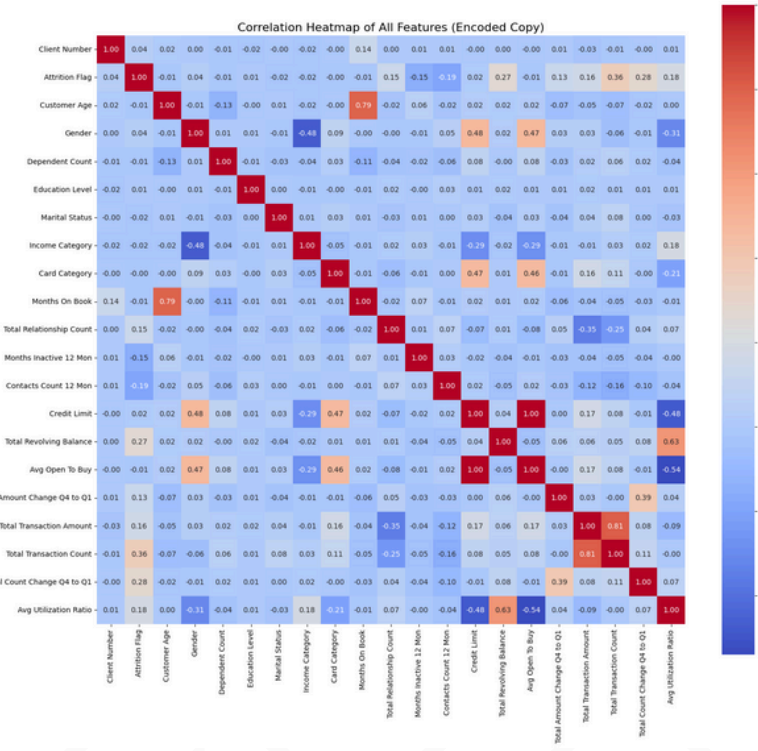
Histogram



Show distribution of values across features

- Low credit utilization rates
- High monthly payment ratio
- Mid-range customer tenure
- Churn follows inactivity trend

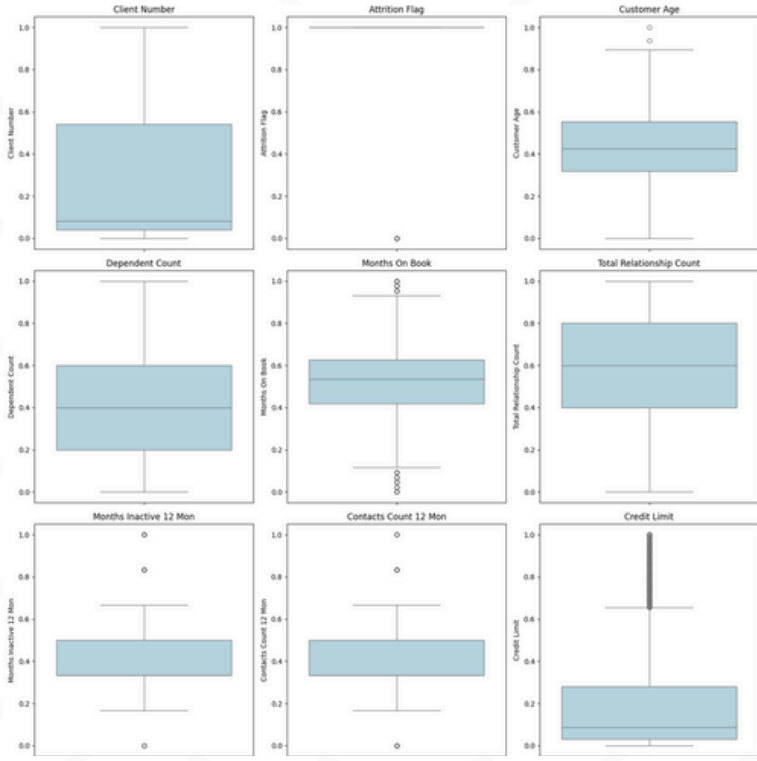
Heatmap



Identify relationships between variables

- Spending count ↔ amount
- Credit limit ↔ open credit
- Age ↔ customer tenure
- Churn = multi-factor outcome

Boxplot



Detect outliers and data spread in numerical features

- Many features show outliers
- Long-tailed financial distributions
- Extreme values = few customers
- Age and tenure more normal

# Dataset Issues & Pre-processing

The dataset had a few issues that needed to be addressed before we proceeded with predictive modelling

## Fixing column names

- Dropped ‘\_’ and changing column names to full term for readability

```
[ 'Client Number',
  'Attrition Flag',
  'Customer Age',
  'Gender',
  'Dependent Count',
  'Education Level',
  'Marital Status',
  'Income Category',
  'Card Category',
  'Months On Book',
  'Total Relationship Count',
  'Months Inactive 12 Mon',
  'Contacts Count 12 Mon',
  'Credit Limit',
  'Total Revolving Balance',
  'Avg Open To Buy',
  'Total Amount Change Q4 to Q1',
  'Total Transaction Amount',
  'Total Transaction Count',
  'Total Count Change Q4 to Q1',
  'Avg Utilization Ratio']
```

## Hidden Missing Values

- Turned ‘Unknown’ strings in categorical data into NaN values to drop easier

```
# -----
# 2.0 DATA CLEANING
# -----

# 2.1 Handling Missing Values

# Dataset has no 'missing values' but the initial data overview indicates some features have 'unknown'
# After looking through the raw dataset, there is a total of 3,380 'Unknown'. These are not recognized

# Convert all unknown in the dataset to NaN values
df.replace('Unknown', np.nan, inplace=True)

# Checking if values were successfully converted to NaN values
display(df.head())
```

|   | CLIENTNUM | Attrition_Flag    | Customer_Age | Gender | Dependent_count | Education_Level | Marital_Status |
|---|-----------|-------------------|--------------|--------|-----------------|-----------------|----------------|
| 0 | 768805383 | Existing Customer | 45           | M      | 3               | High School     | Married        |
| 1 | 818770008 | Existing Customer | 49           | F      | 5               | Graduate        | Single         |
| 2 | 713982108 | Existing Customer | 51           | M      | 3               | Graduate        | Married        |
| 3 | 769911858 | Existing Customer | 40           | F      | 4               | High School     | NaN            |
| 4 | 709106358 | Existing Customer | 40           | M      | 3               | Uneducated      | Married        |

## Extreme Outliers

- Handled outliers by capping values in key numerical columns at the 1st and 99th percentiles to reduce bias

```
<class 'pandas.core.frame.DataFrame'>
Index: 7081 entries, 0 to 10126
Data columns (total 21 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Client Number                        7081 non-null   float64
1   Attrition Flag                       7081 non-null   int64
2   Customer Age                         7081 non-null   float64
3   Gender                               7081 non-null   int64
4   Dependent Count                     7081 non-null   float64
5   Education Level                     7081 non-null   int64
6   Marital Status                      7081 non-null   int64
7   Income Category                     7081 non-null   int64
8   Card Category                       7081 non-null   int64
9   Months On Book                      7081 non-null   float64
10  Total Relationship Count             7081 non-null   float64
11  Months Inactive 12 Mon              7081 non-null   float64
12  Contacts Count 12 Mon               7081 non-null   float64
13  Credit Limit                        7081 non-null   float64
14  Total Revolving Balance             7081 non-null   float64
15  Avg Open To Buy                    7081 non-null   float64
16  Total Amount Change Q4 to Q1       7081 non-null   float64
17  Total Transaction Amount            7081 non-null   float64
18  Total Transaction Count             7081 non-null   float64
19  Total Count Change Q4 to Q1       7081 non-null   float64
20  Avg Utilization Ratio               7081 non-null   float64
dtypes: float64(15), int64(6)
memory usage: 1.2 MB
```

## Categorical Data

- Encoded categorical data using LabelEncoder() as ML require numerical input

```
# Encoding target categorical variable
le = LabelEncoder()
df['Attrition_Flag'] = le.fit_transform(df['Attrition_Flag']) # 'Attrited Customer' = 0, 'Existing Customer' = 1

# Encoding remaining categorical features before feature selection
for col in df.columns:
    if df[col].dtype == 'object':
        le = LabelEncoder()
        df[col] = le.fit_transform(df[col])
```

# Dataset Issues & Pre-processing

## MinMax Scaling

| --- Data After Transformation --- |                 |                         |                        |          |              |         |
|-----------------------------------|-----------------|-------------------------|------------------------|----------|--------------|---------|
|                                   | Client Number   | Attrition Flag          | Customer Age           | Gender   | Dependent    | Count \ |
| 0                                 | 0.505115        | 1                       | 0.404255               | M        |              | 0.6     |
| 1                                 | 0.920736        | 1                       | 0.489362               | F        |              | 1.0     |
| 2                                 | 0.049078        | 1                       | 0.531915               | M        |              | 0.6     |
| 4                                 | 0.008520        | 1                       | 0.297872               | M        |              | 0.6     |
| 5                                 | 0.041421        | 1                       | 0.382979               | M        |              | 0.4     |
|                                   | Education Level | Marital Status          | Income Category        | Card     | Category     | \       |
| 0                                 | High School     | Married                 | \$60K - \$80K          |          | Blue         |         |
| 1                                 | Graduate        | Single                  | Less than \$40K        |          | Blue         |         |
| 2                                 | Graduate        | Married                 | \$80K - \$120K         |          | Blue         |         |
| 4                                 | Uneducated      | Married                 | \$60K - \$80K          |          | Blue         |         |
| 5                                 | Graduate        | Married                 | \$40K - \$60K          |          | Blue         |         |
|                                   | Months On Book  | ...                     | Months Inactive 12 Mon | Contacts | Count 12 Mon | \       |
| 0                                 | 0.604651        | ...                     | 0.166667               |          | 0.500000     |         |
| 1                                 | 0.720930        | ...                     | 0.166667               |          | 0.333333     |         |
| 2                                 | 0.534884        | ...                     | 0.166667               |          | 0.000000     |         |
| 4                                 | 0.186047        | ...                     | 0.166667               |          | 0.000000     |         |
| 5                                 | 0.534884        | ...                     | 0.166667               |          | 0.333333     |         |
|                                   | Credit Limit    | Total Revolving Balance | Avg Open To Buy        | \        |              |         |
| 0                                 | 0.340190        | 0.308701                | 0.345116               |          |              |         |
| 1                                 | 0.206112        | 0.343266                | 0.214093               |          |              |         |
| 2                                 | 0.059850        | 0.000000                | 0.098948               |          |              |         |
| 4                                 | 0.099091        | 0.000000                | 0.136557               |          |              |         |
| 5                                 | 0.077747        | 0.495431                | 0.079970               |          |              |         |

## Feature Selection

```
# Step 1: Manual Feature Dropping based on correlation and multicollinearity
columns_to_drop = [
    'Client Number',      # Identifier
    'Education Level',    # Low correlation
    'Marital Status',     # Low correlation
    'Card Category',      # Low correlation
    'Dependent Count',    # Low correlation
    'Avg Open To Buy'     # Perfect correlation with Credit Limit
]
```

Top 5 features selected by SelectKBest:  
['Contacts Count 12 Mon', 'Total Revolving Balance', 'Total Transaction Count',  
'Total Count Change Q4 to Q1', 'Avg Utilization Ratio']

Shape before dropping columns: (7081, 21)  
Shape after dropping columns: (7081, 15)

# Handling Imbalanced Dataset

There are **significantly fewer attritted customers** compared to existing ones.

## SMOTE (Synthetic Minority Over-sampling Technique)

- Creates synthetic examples of the minority class
- Helps model learn patterns from limited churn data
- Reduces bias toward majority class
- Improves recall and F1-score for minority class

## Class Weighting

- Assigns higher penalty for misclassifying churn
- No new data created — adjusts loss function during training
- Simple and effective for many models
- Often combined with SMOTE for best results



# Model Selection

## Logistic Regression

- Simple, fast, and easy to interpret
- Good baseline model for binary classification
- Performs well when relationship is linear

## Random Forest

- Ensemble of decision trees (bagging)
- Handles non-linearity and feature interactions
- Robust to noise and outliers
- Built-in feature importance ranking

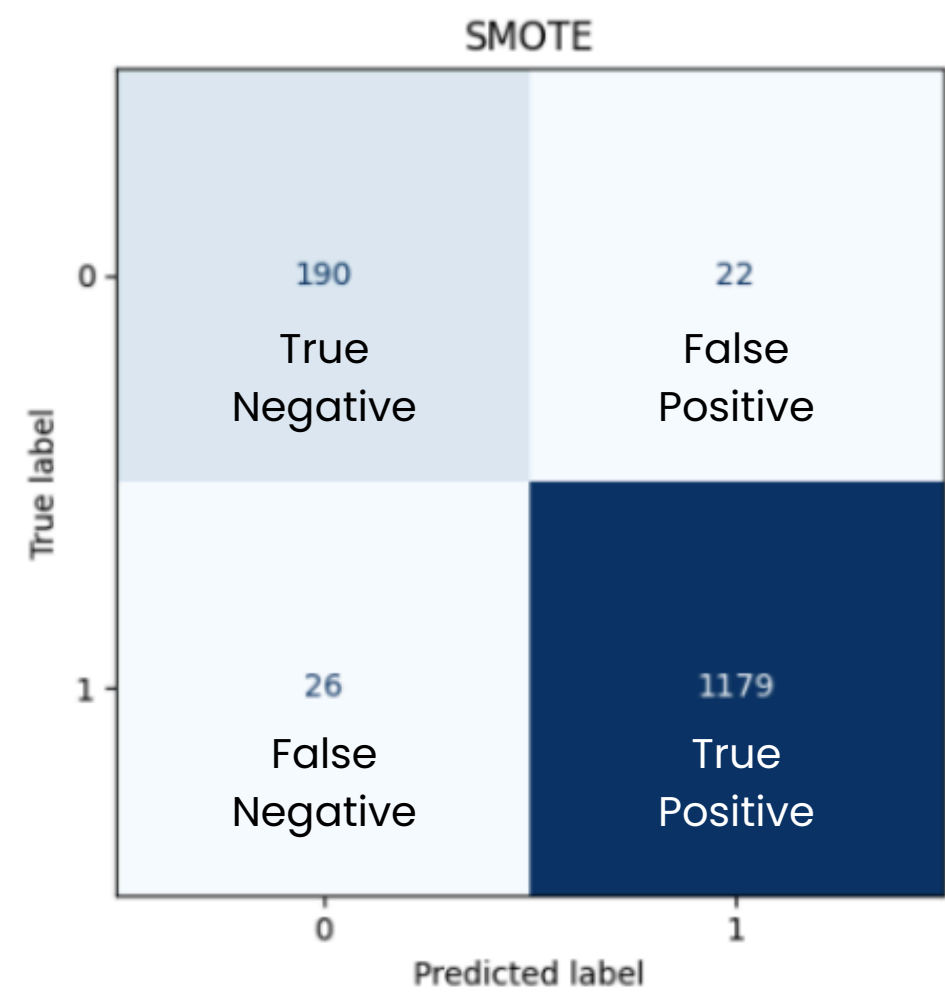
## XGBoost

- Gradient boosting framework with high predictive power
- Captures complex patterns and feature interactions
- Excellent with structured/tabular data

# Top 3 Class 0 Recall in Confusion Matrix

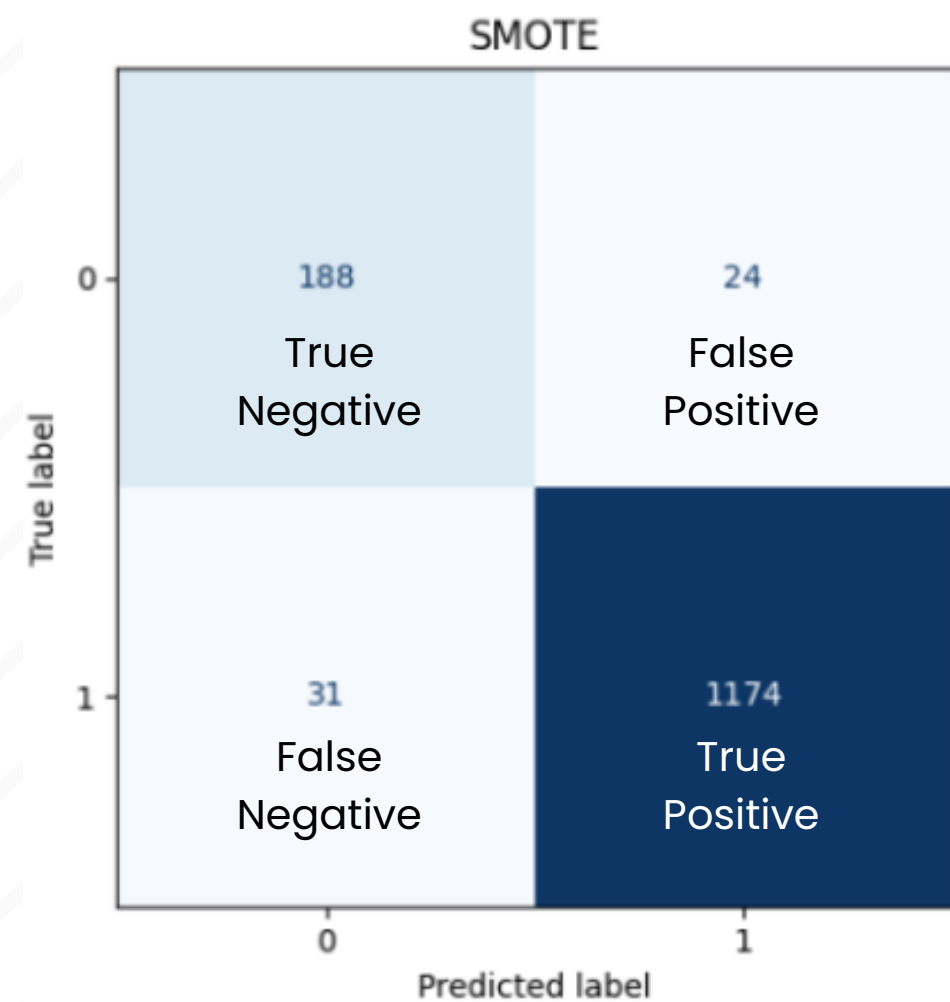
Pre Stratified K-Fold + Cross Validation

## 1. XGBoost (SMOTE)



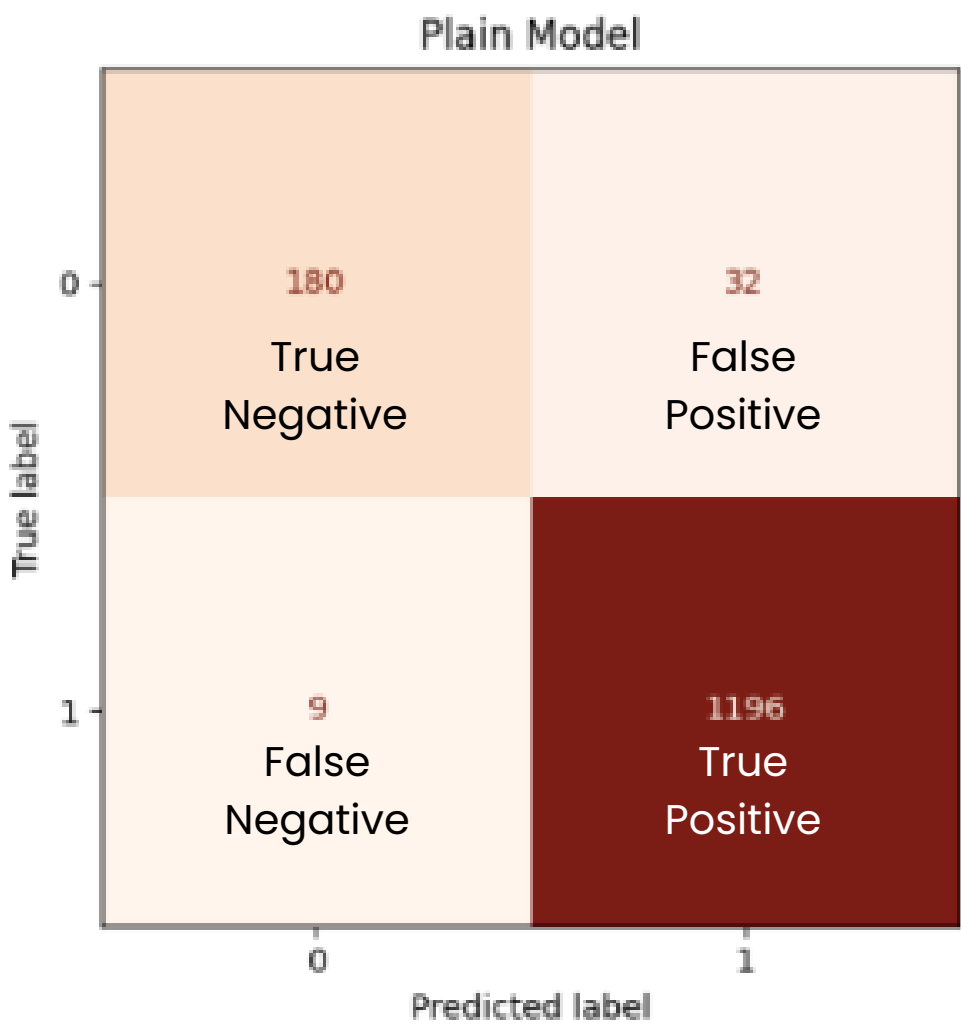
Class 0 Precision: 87.98%  
Class 0 Recall: 89.61%

## 2. Random Forest (SMOTE)



Class 0 Precision: 85.83%  
Class 0 Recall: 88.68%

## 3. XGBoost (Plain)



Class 0 Precision: 95.24%  
Class 0 Recall: 84.11%



# Top 3 Class 0 Recall Across All Models

Post Stratified K-Fold + Cross Validation

## 1. Weighted XGBoost

| Metric                | XGB - Weighted |
|-----------------------|----------------|
| Test Accuracy         | 0.965541       |
| Test F1 Macro         | 0.936334       |
| Test Recall (Class 0) | 0.916422       |
| Test Recall (Class 1) | 0.974699       |
| Test F1 (Class 0)     | 0.893213       |

## 2. Random Forest with SMOTE

| Metric                | RF - SMOTE |
|-----------------------|------------|
| Test Accuracy         | 0.95735    |
| Test F1 Macro         | 0.920326   |
| Test Recall (Class 0) | 0.875999   |
| Test Recall (Class 1) | 0.972519   |
| Test F1 (Class 0)     | 0.866015   |

## 3. Weighted Logistic Regression

| Metric                | LogReg - Class Weight |
|-----------------------|-----------------------|
| Test Accuracy         | 0.834203              |
| Test F1 Macro         | 0.752415              |
| Test Recall (Class 0) | 0.824829              |
| Test Recall (Class 1) | 0.835956              |
| Test F1 (Class 0)     | 0.610132              |

# Final Model Selection

## 1. Weighted XGBoost

For Banks, it is often more costlier to let churners slip through the cracks, Weighted XGBoost boasts the Strongest Recall of all models with consistent Accuracy overall after testing through Stratified K-Fold and Cross Validation. Whereas other model falls short with noticable decrease in performance.

### Top 1

Class 0 Recall  
**91.64%**

Class 0 F1  
**89.32%**

Accuracy  
**96.55%**

### Feature Importance (Gain)

1. **Total Transaction Count (71.88)**
2. **Total Revolving Balance (38.16)**
3. **Total Transaction Amount (30.31)**

\*Measured by Gain-Based Importance Score



# Conclusion

We successfully achieved our project goal of building accurate churn prediction models and uncovering key behavioral insights to support customer retention strategies.

## Key Churn Predictors

### Common predictive features

- Total Transaction Count
- Total Transaction Amount
- Total Count Change Q4 to Q1
- Total Revolving Balance
- Change in Transactions Count (Q4–Q1)
- Months Inactive in Last 12 Months
- Number of Bank Contacts

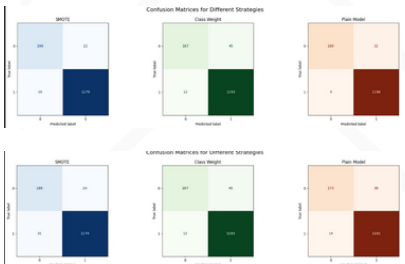
Less engagement and **lower financial activity** are likely to predict churn

**Monitor** Customer financial behavior for effective retention strategies

## Model Accuracy

### Model Selection

- Logistics Regression
- Random Forest
- XGBoost

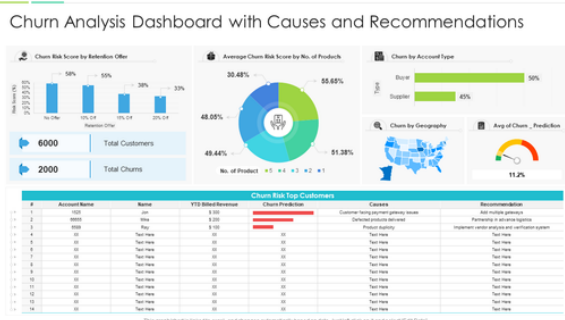


Logistic: ~79.2% accuracy  
Random: ~82.4% accuracy  
XGBoost: **84.1% accuracy**

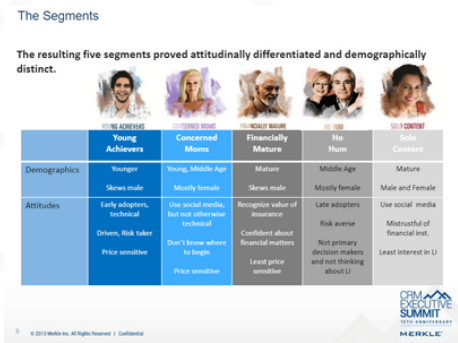
**XGBoost** with class weighting delivers reliable 80%+ accuracy for churn prediction

## Strategies for Banks to Reduce Churn

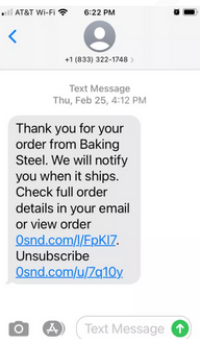
### Early Warning Systems



### Campaigns for Segmented Retention



### Proactive Personal Engagement



Implementing these strategies to **strengthen** customer retention and **reduce** churn in banks